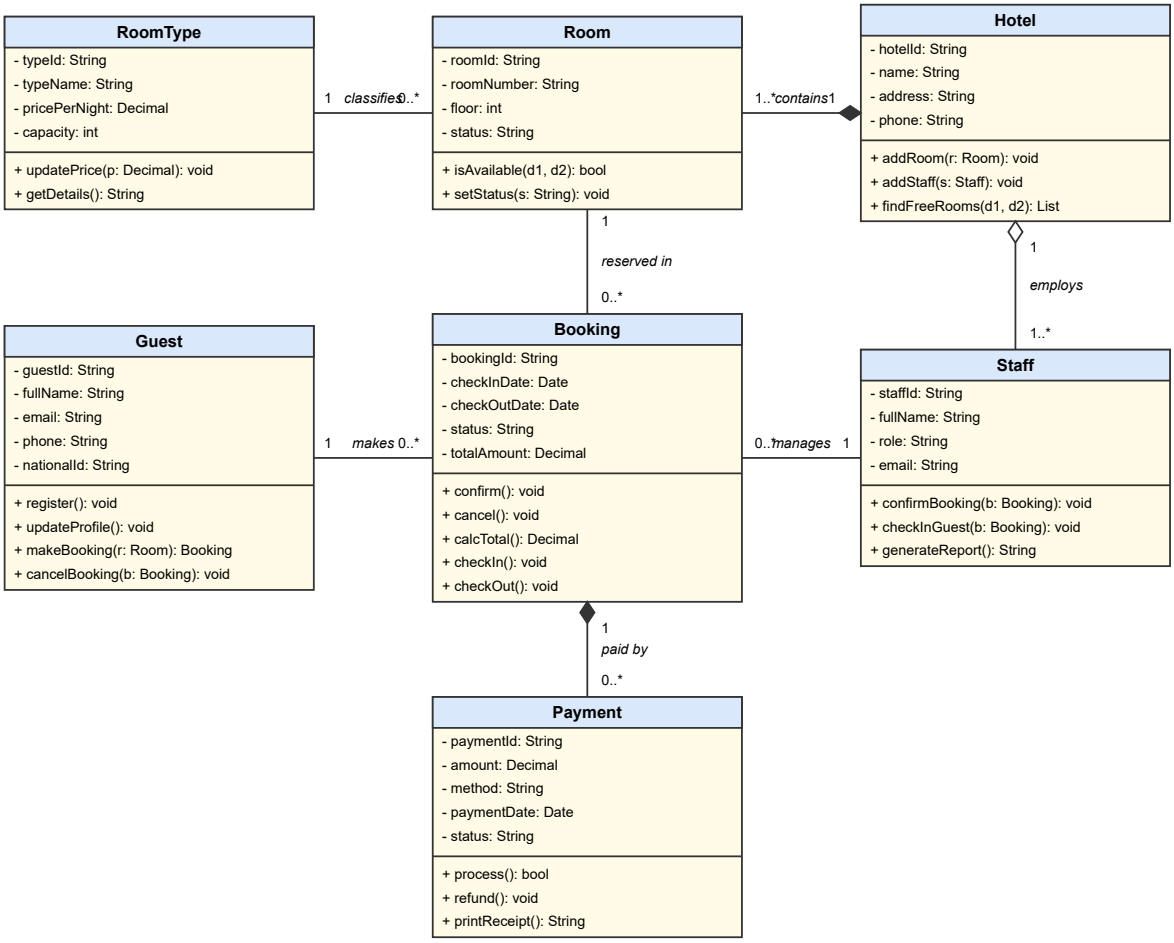
Report 2: UML Class Diagram Report

Course: UML Class Diagram Modelling | Instructor: Eric Maniraguha | Institution: Kigali Independent University (ULK)
Group: Mugisha Alain Paisible & MUCUNGUZI Felix | Date: 20 October 2026

1. UML Class Diagram



2. Classes, Attributes and Methods

Class	Attributes	Methods
Hotel	- hotelId: String - name: String - address: String - phone: String	+ addRoom(r: Room): void + addStaff(s: Staff): void + findFreeRooms(d1, d2): List
RoomType	- typeId: String - typeName: String - pricePerNight: Decimal - capacity: int	+ updatePrice(p: Decimal): void + getDetails(): String
Room	- roomId: String - roomNumber: String	+ isAvailable(d1, d2): bool + setStatus(s: String): void

	- floor: int - status: String	
Guest	- guestId: String - fullName: String - email: String - phone: String - nationalId: String	+ register(): void + updateProfile(): void + makeBooking(r: Room): Booking + cancelBooking(b: Booking): void
Booking	- bookingId: String - checkInDate: Date - checkOutDate: Date - status: String - totalAmount: Decimal	+ confirm(): void + cancel(): void + calcTotal(): Decimal + checkIn(): void + checkOut(): void
Staff	- staffId: String - fullName: String - role: String - email: String	+ confirmBooking(b: Booking): void + checkInGuest(b: Booking): void + generateReport(): String
Payment	- paymentId: String - amount: Decimal - method: String - paymentDate: Date - status: String	+ process(): bool + refund(): void + printReceipt(): String

3. Relationships and Multiplicities

From	To	Name	Type	Multiplicity (from : to)	Explanation
Hotel	Room	contains	comp	1 : 1..*	Composition: a hotel contains one or more rooms; rooms cannot exist without the hotel.
RoomType	Room	classifies	assoc	1 : 0..*	Association: each room has exactly one room type; a type can apply to many rooms.
Guest	Booking	makes	assoc	1 : 0..*	Association: a guest can make many bookings; each booking belongs to one guest.
Room	Booking	reserved in	assoc	1 : 0..*	Association: a room can be reserved in many bookings over time; each booking is for one room.
Booking	Payment	paid by	comp	1 : 0..*	Composition: payments belong to a booking and are removed with it; a booking may have several part payments.
Hotel	Staff	employs	aggr	1 : 1..*	Aggregation: a hotel employs one or more staff; staff can exist as records independently.
Staff	Booking	manages	assoc	1 : 0..*	Association: a staff member manages many bookings; each booking is handled by one staff member.

4. Summary

A **Guest** makes many **Bookings**. Each Booking reserves one **Room**, is handled by one **Staff** member and is paid through one or more **Payments**. Rooms belong to a **Hotel** and are classified by a **RoomType**. Composition is used where the part cannot live without the whole (Hotel-Room, Booking-Payment), aggregation where it can (Hotel-Staff).